

# HLD Case Study 2: Ticketing and Hotel Reservation (Ticketmaster / Booking.com)

**Key Focus Areas:** Preventing double booking, temporary holds with expiry, flash-sale traffic spikes, inventory consistency, and payment integration.

## 1. Requirements and Scope

**Functional:** search events or hotels, view availability, select seats or room dates, **hold** inventory for a few minutes, pay, receive confirmation, cancel/refund. **Non-functional:** **never sell the same seat or room-night twice** (strong consistency on inventory); browsing may be eventually consistent; survive a 100x spike when a popular event goes on sale; hold expiry must be reliable. **Out of scope:** dynamic pricing algorithms, resale marketplace, recommendations.

## 2. Estimates

Assume a major on-sale: 50,000 seats, 2M users arriving in the first 10 minutes, each making about 20 requests (seat map refreshes, holds).

The write path is small (about 3.3K/s) but **extremely contended**: 120 users compete for each good seat. The design goal is correct arbitration under contention, plus shielding the database from the read flood.

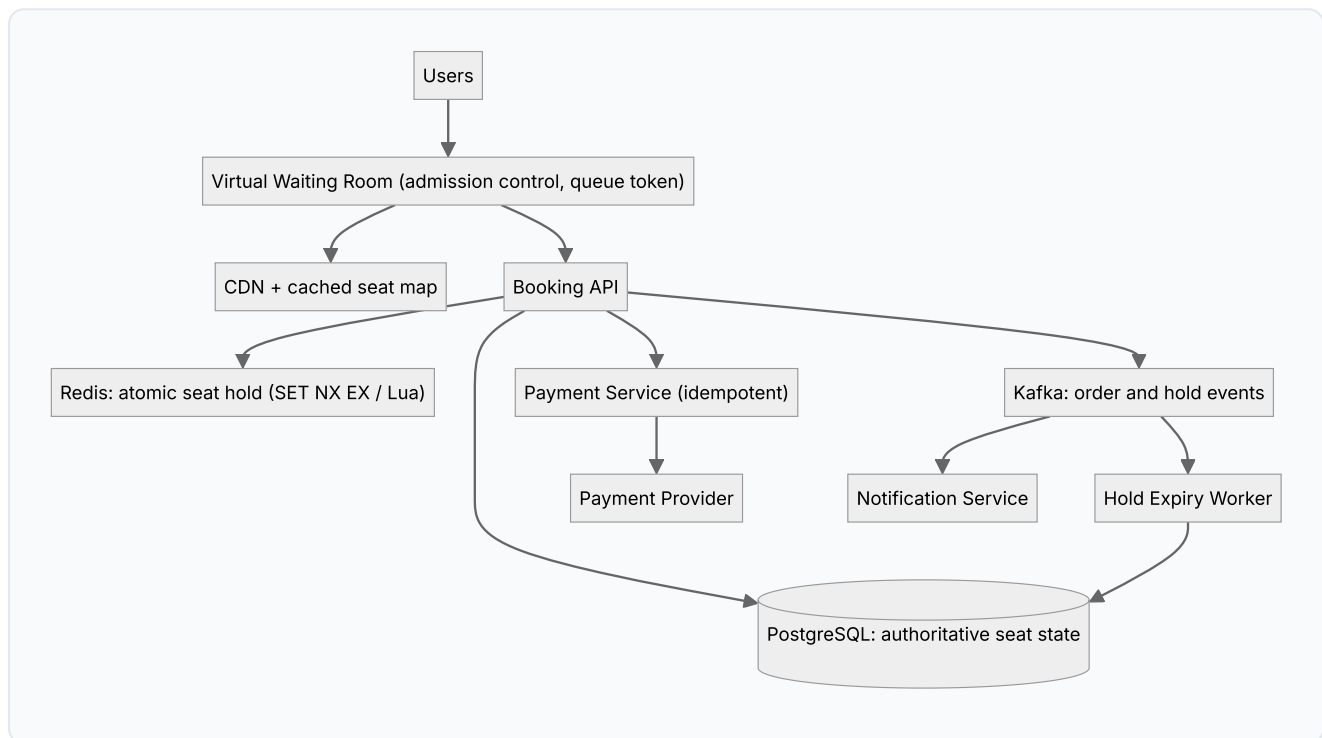
## 3. API Design

- `GET /events/{id}/seatmap` (cached, may be a few seconds stale)
- `POST /holds` body `{event_id, seat_ids[], idempotency_key}` returns `{hold_id, expires_at}` or `409 Conflict`
- `POST /orders` body `{hold_id, payment_token, idempotency_key}` returns `{order_id, status}`
- `DELETE /holds/{hold_id}` (user abandons)

## 4. Data Model

- **seat** ( `event_id`, `seat_id`, `status`: AVAILABLE|HELD|SOLD, `hold_id`, `hold_expires_at`, `version` ) in a relational database (PostgreSQL), partitioned by `event_id`.
- **hold** ( `hold_id`, `user_id`, `seat_ids`, `expires_at` ), **order** ( `order_id`, `hold_id`, `amount`, `payment_status` ).
- For hotels the unit is a **room-night**: `inventory`(`hotel_id`, `room_type`, `date`, `total`, `reserved`) with a check `reserved ≤ total`.

## 5. Architecture



## 6. Deep Dive: Preventing Double Booking

Three layers, increasing in cost:

1. **Fast arbitration in Redis.** `SET seat:{event}:{seat} {hold_id} NX EX 600` is atomic: exactly one caller gets `OK`. Losers get `409` immediately without touching the database. For multi-seat holds use a Lua script that acquires all seats or none.
2. **Authoritative write in the database.** On successful hold, persist with optimistic concurrency: `UPDATE seat SET status='HELD', hold_id=:h, hold_expires_at=:t, version=version+1 WHERE event_id=:e AND seat_id=:s AND status='AVAILABLE'` and check `rowcount = 1`. Redis is a performance layer; the DB row is the source of truth, so a Redis failure cannot cause double sale.

3. **Final purchase check.** On payment success, `UPDATE ... SET status='SOLD' WHERE hold_id=:h AND status='HELD' AND hold_expires_at > now()`. If zero rows, the hold expired: refund automatically.

**Hold expiry.** Redis TTL frees the fast lock; a worker also scans `hold_expires_at < now()` and resets stale `HELD` rows to `AVAILABLE` (idempotent, runs every few seconds). Never rely on a single mechanism.

**Waiting room.** Admit users at the rate the backend can serve (for example 1,000 per second) using signed queue tokens; everyone else sees a lightweight static page from the CDN. This turns an unbounded spike into a controlled flow.

**Hotels.** Decrement inventory per night in one transaction: `UPDATE inventory SET reserved = reserved + 1 WHERE hotel_id=:h AND room_type=:r AND date BETWEEN :in AND :out - 1 AND reserved < total`, and require rowcount equals number of nights, otherwise roll back. Overbooking is sometimes deliberate; model it as `total = physical * 1.05` per business policy.

## 7. Scaling and Bottlenecks

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- **Reads:** cache seat maps at the CDN (1 to 5 s TTL) and show "likely available" rather than exact state.
- **Hot partition:** one blockbuster event is one hot key range. Shard by `(event_id, section)` so sections lock independently.
- **Database write contention:** the Redis gate keeps conflicting writers out of the DB, so only about one write per successful hold arrives.
- **Payment latency (seconds):** never hold a DB transaction open during payment; hold state is separate from the payment call.

## 8. Failure Modes and Mitigations

| Failure                                | Mitigation                                                                                |
|----------------------------------------|-------------------------------------------------------------------------------------------|
| Redis loses data                       | DB constraint still prevents double sale; rebuild holds from DB<br><code>HELD</code> rows |
| Payment succeeds but callback lost     | idempotency key + reconciliation job against the PSP                                      |
| User pays after hold expiry            | conditional <code>UPDATE</code> fails, trigger refund                                     |
| Duplicate submit (double click, retry) | <code>idempotency_key</code> returns the same result                                      |
| Bots grabbing seats                    | waiting room, CAPTCHA, per-account and per-IP limits, signed tokens                       |
| Region outage                          | active-passive DB with synchronous replica for seat state                                 |

## 9. Trade-offs

- **Pessimistic lock vs optimistic version check:** pessimistic ( `SELECT ... FOR UPDATE` ) is simple but blocks under contention; optimistic fails fast and retries, which suits "first wins" semantics.
- **Redis-first vs DB-only:** Redis cuts DB load by about 20x under contention but adds a second system; keep DB authoritative.
- **Strong vs eventual consistency:** strong for inventory, eventual for browse and search results.
- **Short vs long holds:** longer holds improve conversion but reduce inventory turnover; 5 to 10 minutes is typical.

## 10. Interview Timeline and Follow-ups

Spend 10 minutes on the double-booking argument: state the invariant ("a seat is SOLD at most once"), show the conditional update, then add Redis as an optimisation. **Follow-ups:** How do you handle best-available seating? (query candidates, try holds in order, release on partial failure). How do refunds restore inventory? (event-driven, idempotent). How would you add dynamic pricing? (pricing service reads demand metrics; price is snapshotted on the hold). What if the waiting room itself is attacked? (CDN rate limits, token signing, proof-of-work).